

## Project proposal feedback

**Project title:** Knowledge Graph Link Prediction with Deep Knowledge Graph Embedding Models **Group:** Thalia Delhaise, Lenny Briclet, André Fonseca, Rafael Gufler

## Overall assessment

This is a promising proposal. The problem is clear, the dataset is public and standard, the methodological path is credible, and the evaluation protocol is much more precise than in many early proposals. The project is also well aligned with the course requirement of using real data and producing a concrete experimental pipeline.

The main issue: the proposal is currently framed mostly as a **baseline comparison on a standard benchmark**. That is valid, but it would become stronger if it were sharpened into a more explicit experimental question rather than “train TransE and RotatE and compare them.”, especially considering that the group is composed by 4 members.

## What is already good

- The problem is well defined: link prediction in knowledge graphs using triples  $((h,r,t))$ , with clear train/validation/test separation.
- The dataset choice is good: **FB15k-237** is public, standard, and of manageable scale for a course project.
- The proposal already specifies a realistic pipeline: ingest triples, map entities and relations, train on the official split, and evaluate on held-out triples.
- The method plan is credible: **TransE** as baseline and **RotatE** as stronger model, both through PyKEEN, is a sound starting point.
- The evaluation protocol is unusually clear for a first proposal: filtered ranking, MRR, Hits@1/3/10, and qualitative inspection of predictions.
- The proposal is already close to what the project brief asks for: problem, motivation, data, method, and evaluation are all present.

## Main issues to address

- Please let me know via email if all members are doing the KG course.
- The project is currently a bit too close to a clean benchmark reproduction. It needs a sharper experimental question to gain scientific depth.
- The method section is sound, but it relies heavily on **PyKEEN**. That is fine, but the group should make sure the work is not reduced to calling a library with default settings. Make clear what part of the work is the group’s own contribution beyond using PyKEEN.
- The proposal does not yet say what the **main claim** is. What exactly will be learned from the comparison between models?

## Detailed feedback

### 1. Problem definition

The problem is clearly stated and technically precise. The proposal explains what a knowledge graph is, what link prediction means, and what score function is to be learned. This is strong. I can immediately see the task and the intended training and evaluation pipeline.

What is still missing is a more explicit **research question**. Right now, the proposal describes a task and a pipeline, but not yet a clear question such as: “Which relation patterns benefit more from RotatE than from TransE?” or “How much does increased expressive power improve filtered ranking on FB15k-237 under equal training conditions?” That small change would make the project more scientific.

### 2. Data

The data choice is very good. FB15k-237 is public, established, already split, and large enough to be meaningful while still feasible for a course project. It avoids the common risks of collecting private data, creating labels from scratch, or spending too much time on preprocessing.

A good next step is to briefly characterize the dataset in Milestone 1 beyond the official counts: for example, relation frequency distribution, degree distribution, and examples of one-to-one, one-to-many, many-to-one, and many-to-many relations. That would help motivate later error analysis.

### 3. Method and baseline

The baseline choice is appropriate. **TransE** is a classical and interpretable starting point; **RotatE** is a sensible stronger model. Using the same splits and evaluator through PyKEEN is also good for fairness.

However, the project should avoid becoming only “run two standard models.” To strengthen the methodological contribution, I suggest defining one controlled comparison axis. For example:

- embedding dimension
- number of negative samples
- loss function or margin
- early stopping criterion
- relation-type analysis
- effect of inverse relations or filtered protocol details

Even a modest but well-motivated ablation would make the project much stronger.

#### 4. Evaluation

This is one of the strongest parts of the proposal. The use of **filtered MRR** and **Hits@1/3/10** is appropriate for knowledge graph link prediction, and the proposal already distinguishes head and tail prediction.

Still, the evaluation can be improved in two ways. First, add **relation-wise analysis** rather than only aggregate metrics. Second, include a small **error analysis** section: inspect a few successes and failures, and try to explain them in terms of relation structure, ambiguity, or entity frequency. That would give the report more depth than a benchmark table alone.

#### 5. Feasibility

This project looks **feasible** within the course constraints.

#### 6. Experimental depth

At the moment, the project is solid but slightly conservative. It is not shallow in the sense of being trivial, but it is at risk of becoming mostly a standard reproduction with clean reporting.

To deepen it without making it risky, choose one or two of the following directions:

- compare TransE and RotatE by **relation pattern** rather than only overall score
- study the effect of **embedding dimension / training budget** on performance-efficiency trade-offs
- analyze how performance changes for **frequent vs rare entities/relations**
- compare **filtered vs raw ranking** conceptually and discuss why filtered evaluation matters
- add a simple but meaningful **ablation** on negative sampling or loss configuration

Any one of these would already make the project more interesting. The goal is to have a project that is small enough to finish, but deep enough to learn something real.

#### Suggested minimum plan for success

By **Milestone 1**, the group should already have:

- the dataset downloaded and parsed
- entity and relation mappings implemented or verified
- one baseline running end-to-end
- preliminary training curves
- one evaluation script producing MRR and Hits@K
- a short list of the final experiments they will run

By the **final report**, a solid version of the project should include:

- one clear task on one main dataset
- TransE and RotatE under matched conditions
- one additional experimental axis or ablation
- aggregate metrics plus relation-wise analysis
- a short qualitative/error analysis section
- a clear discussion of compute time and practical trade-offs

## conclusions

### How to improve the proposal

- Change the title to something more specific
- Rewrite the objective as a **specific experimental question**, not only as “perform link prediction with knowledge graph embeddings.”
- Keep **TransE and RotatE** as the core models, but add **one controlled ablation or analysis axis**.
- Make clear what part of the work is the group’s own contribution beyond using PyKEEN.
- Plan from the start to include **relation-wise and error analysis**, not only aggregate metrics.

Congratulations.